



Technical Service Bulletin

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Cylinder Head Reuse for Update from 13-Piece to Single-Piece Cylinder Head Gasket

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Warranty Statement

The information in this document has no effect on present warranty coverage or repair practices, nor does it authorize TRP or Campaign actions.

Contents

This Service/Parts Topic documents specific criteria for the Signature, ISX, and QSX15 Cummins engines built before September, 2001, equipped with a 13-piece head gasket and grooved liners. Cylinder Head Procedure 002-004 and Cylinder Head Gasket Procedure 002-021 in the Signature, ISX, and QSX15 Service Manual, Bulletin 3666239, are being revised to include this information. Recent testing has determined that upgrading to a one-piece head gasket and flattop liners is more resistant to combustion leaks than the 13-piece head gasket and grooved liners. The repair is especially applicable for customers who have had head gasket failures on engines which are **not** ready or scheduled for overhaul.

This information addresses coolant leaks, caused by combustion gases being pushed into the coolant passages or during repairs in which out-of-specification liner recession is measured. Cummins Distributors and Dealers are encouraged to use their technical expertise to implement these guidelines for low cost, high quality repair of Cummins products.

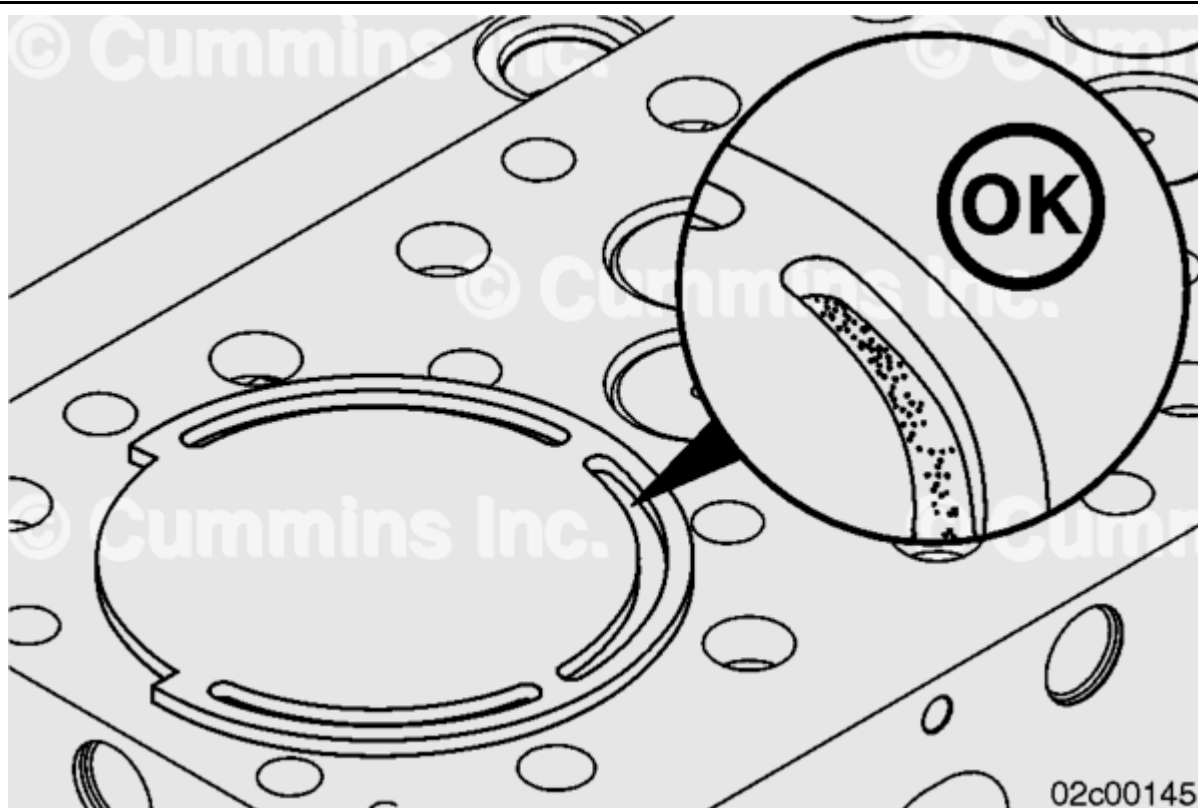
This change in the 'Reuse Guidelines' does **not** apply to instances when, after the cylinder head has been cleaned, there is detectable depth across the sealing area between two cylinders or between a cylinder coolant passage and a capscrew hole.

This change to the reuse guidelines **only** applies to the ESN range before 14022616. Based on the results of Service Engineering analysis and endurance testing, upgrading to a one-piece head gasket and flattop liners reduces susceptibility to coolant leaks. Wear in the area where the fire ring combustion seal contacts the cylinder head is now acceptable when replacing a 13-piece cylinder head gasket and grooved liners with a one-piece cylinder head gasket and flattop liners. To prevent possible damage to the cylinder head during the repair, proper cleaning procedures and all other reuse guidelines **must** be checked and followed as stated in the Cylinder Head Procedure 002-004, in the Signature, ISX, and QSX15 Service Manual, Bulletin 3666239.

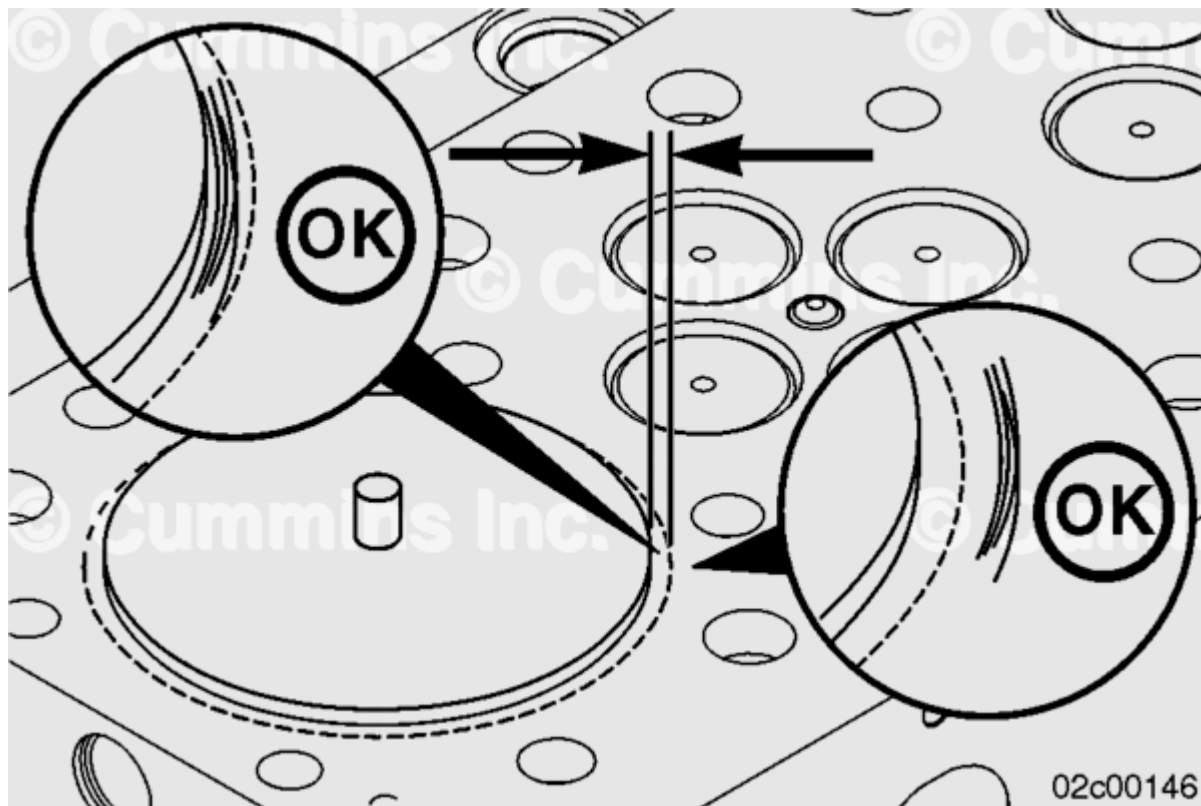
This information applies specifically to ESN 14022616 and earlier, and addresses reuse due to erosion in the sealing area around the combustion surface. For engines outside of this ESN range, Cummins Distributors and Dealers are encouraged to continue to follow the 'Inspect for Reuse' steps in Cylinder Head Procedure 002-004 in the Signature, ISX, and QSX15 Service Manual, Bulletin 3666239

Table 1, Cylinder Head Gasket Part Number Information

Description	Quantity	Old Part Number (13-Piece Gasket with Grooved Liners)	New Part Number (One-Piece Gasket with Flattop Liners)
Gasket Set	1	4025287	4025300
Liner Kit	6	3800453	4089153
Ring Set	6	4089154	4089154



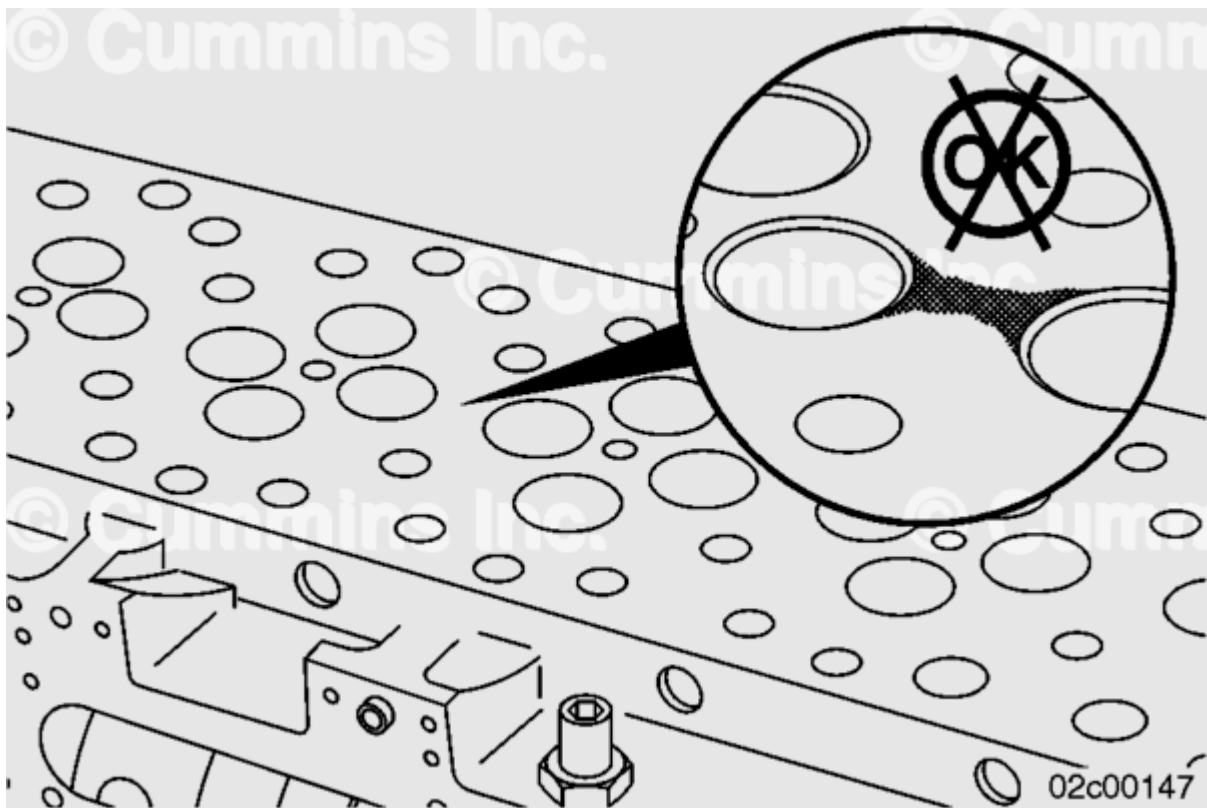
Service Tool Part number 3165071



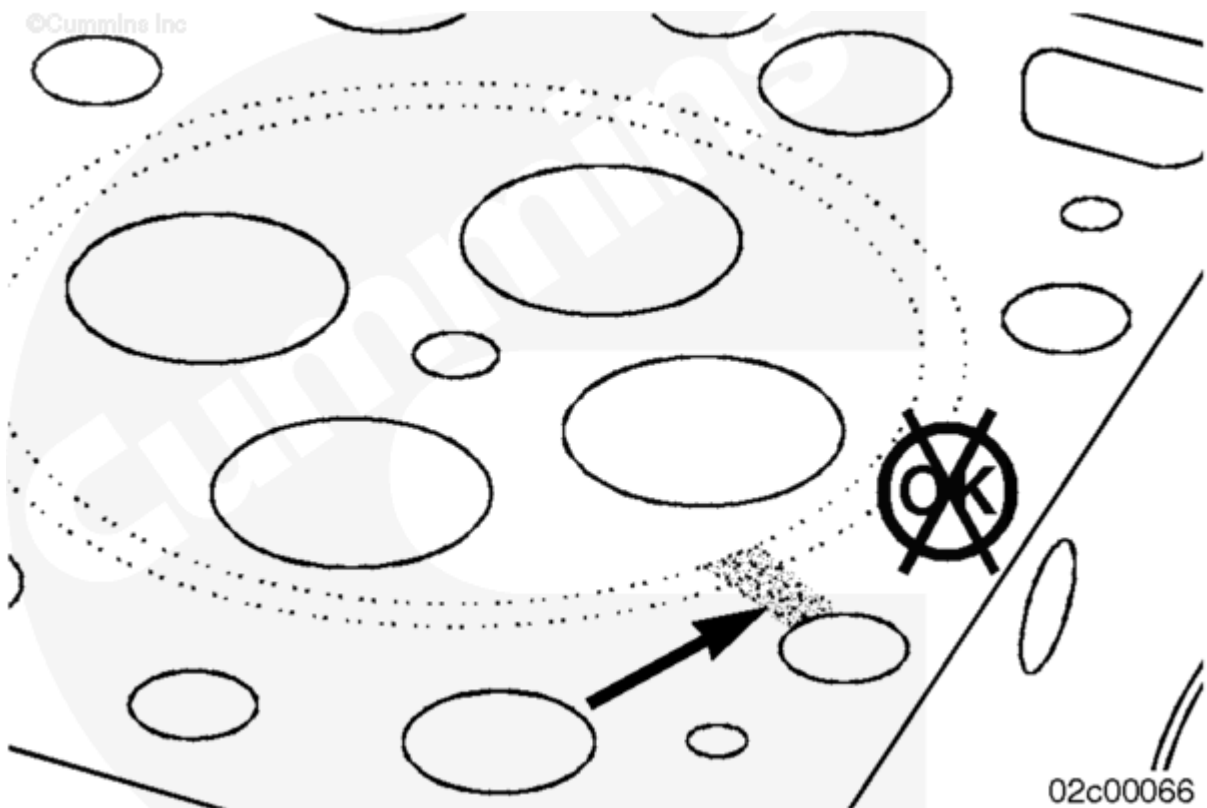
Service Tool Part number 3164442

Note : Wear in the area that the fire ring combustion seal contacts the head is now acceptable when replacing a 13-piece head gasket and grooved liners with a one-piece head gasket and flattop liners.

This change in Reuse Guidelines does **not** apply to instances when, after the cylinder head has been cleaned, there is detectable depth across the sealing area between two cylinders or between a cylinder and coolant passage or a cylinder and capscrew hole.



Depth Across Sealing Area



Leak Path Between Cylinder and Cylinder Head Capscrew Hole

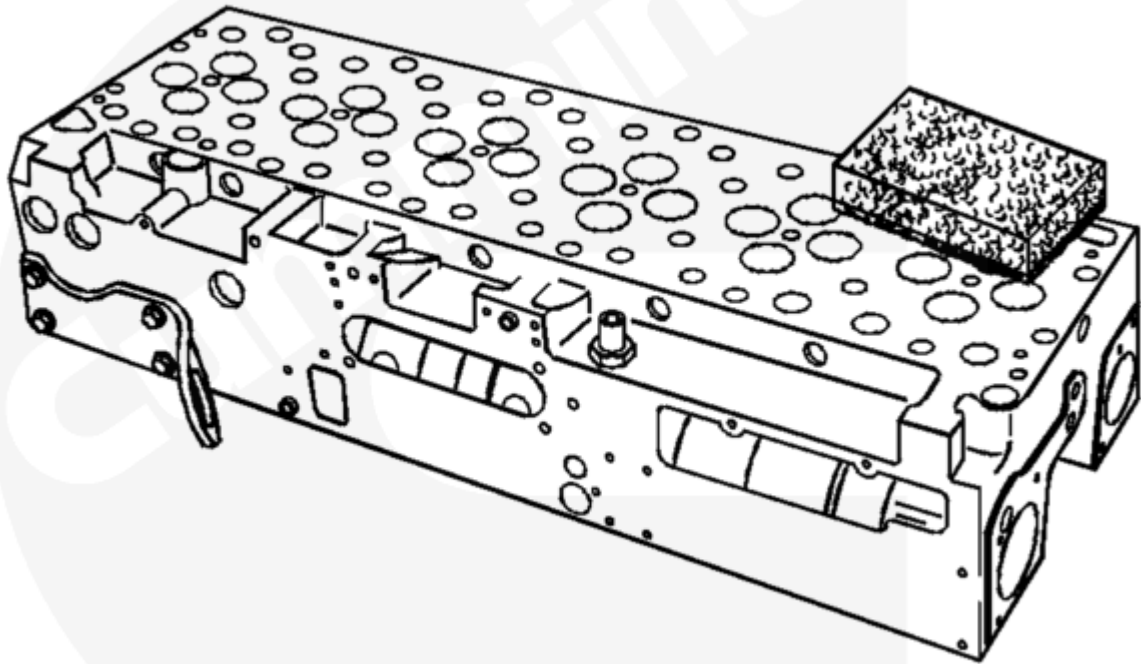
⚠ CAUTION ⚠

Failure to insert the oil passage plugs can result in a connecting rod bearing failure, main bearing failure, or both.

Immediately upon removal of the cylinder head, plugs **must** be installed into the oil passage drillings in both the cylinder head and the block. The plugs are necessary to prevent debris from entering the lubrication system during the repair.

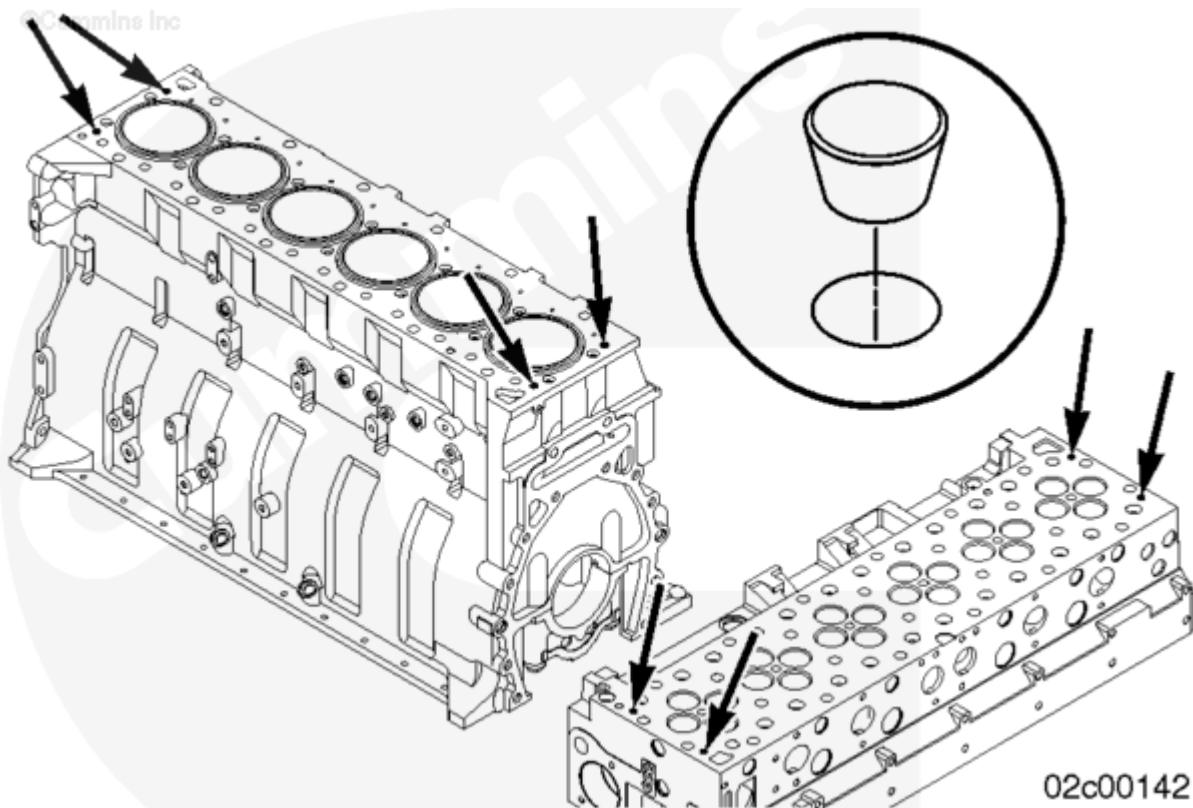
Prior to cleaning the combustion face of the cylinder head, small rubber plugs, Part number 4089255, and tape **must** be used to plug or cover the oil supply holes at the ends of the cylinder head to prevent debris from entering the oil passages.

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Clean Cylinder Head



Protective Plug Locations

Use Scotch-Brite™ 7448 abrasive hand pad, Part Number 3823258, or equivalent, to clean the cylinder head combustion face. Also make sure there is no loose debris sitting in the valve pockets around the head of the valve.

Do **not** use power tools to clean the combustion face of the cylinder head.

For more information on the Protective Plug Kit, consult the Remove step in the Cylinder Head Procedure 002-004, in the Signature, ISX, and QSX15 Service Manual, Bulletin 3666239.

Related Procedures

10-002-004 (/qs3/pubsys2/xml/en/procedures/10/10-002-004.html)	Cylinder Head (/qs3/pubsys2/xml/en/procedures/10/10-002-004.html)
10-002-021 (/qs3/pubsys2/xml/en/procedures/10/10-002-021.html)	Cylinder Head Gasket (/qs3/pubsys2/xml/en/procedures/10/10-002-021.html)